SELECTED SYSTEMS WORK

An AI development team that runs on local hardware

Five plates documenting a multi-agent coding pipeline and the campus AI infrastructure it runs on — every model served on hardware we own.

CONTENTS

MULTI-AGENT CODING PIPELINE

A software team that runs on one desk

Five roles, four local models, no cloud inference. LangGraph orchestration on a single NVIDIA DGX Spark GB10 with 128 GB of unified memory.

---

01 · AGENT

Product Manager

Reads the request and writes the spec: scope, file plan, and the acceptance criteria every other agent is held to.

Output is a JSON contract, not a paragraph of intentions.

MODEL

Nemotron 3 Super · 123B

02 · HUMAN

Tech Lead

Approves or rewrites the spec before a line of code exists.

The graph genuinely blocks here — state is checkpointed and the run resumes on the next human answer.

ROLE

human-in-the-loop gate

03 · AGENT

Developer

Implements against the approved spec. Reads and writes real files through MCP filesystem and git servers.

Retrieval is deterministic: ripgrep and AST, not vectors.

MODEL

Devstral Small 2 · 24B

04 · AGENT

QA Engineer

Runs the suite, parses the failures, and returns a structured verdict rather than a summary of what it assumes probably happened.

Shares the Dev model on purpose — no swap latency.

MODEL

Devstral Small 2 · 24B

05 · AGENT

DevOps

Builds the artifact, previews it, and ships it.

Loop guards cap the retries, detect repeat errors, and escalate instead of spinning on the same failure forever.

MODEL

Gemma4 · 26B

SHARED SERVICES

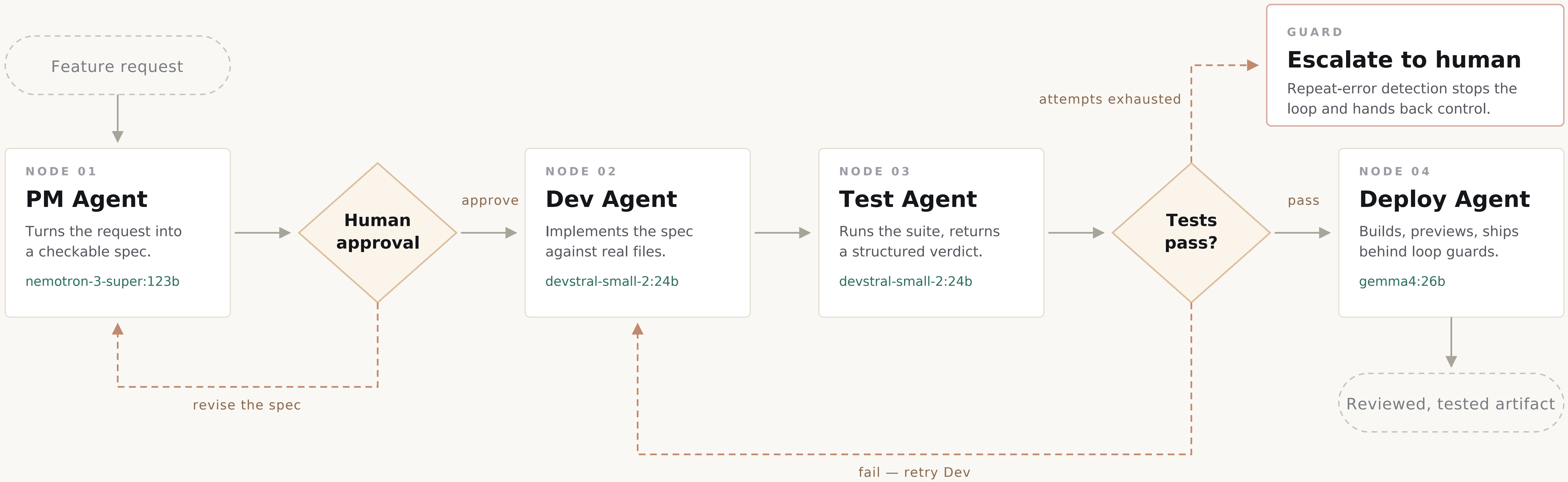
LangGraph state graph · SQLite checkpointing · Langfuse tracing, self-hosted · Ollama runtime · MCP servers: filesystem, git, github, brave-search, playwright, context7

Model provenance restricted to US and EU sources throughout — a hard Texas DIR constraint, not a preference.

EXECUTION GRAPH

How one request becomes shipped code

A LangGraph state machine with a blocking human gate and a conditional retry edge.
Failure is a state the graph knows how to route, not an exception that ends the run.



DURABLE STATE

Every node commits to a SQLite checkpointer. A paused or crashed run resumes at the exact node it left.

OBSERVABILITY

Every model call opens a Langfuse span: prompt, latency, tokens, and which agent asked for what.

TOOL SURFACE

MCP servers give agents real file, git and browser access, filtered per agent at the graph level.

MODEL ROUTING

Dev and Test deliberately share one model so the hot loop never pays a model-swap cost.

IMPLEMENTATION

LangGraph StateGraph with conditional edges · SQLite checkpointer · Ollama serving three models on one GB10 node · Langfuse spans on every call

SYSTEM ARCHITECTURE

One node, seven layers

Everything from the accelerator to the browser tab runs on hardware we control.
No prompt, no file, and no inference request leaves the building.

INTERFACE *where people touch it*	Open WebUI · VS Code + Continue.dev · Claude Code
ORCHESTRATION *the state machine*	LangGraph state graph · SQLite checkpointer · Human approval gate · Conditional retry edge
AGENTS *four roles, one graph*	PM · Dev · Test · Deploy · SkillLoader → per-agent skills
TOOLING *real repo access*	filesystem · git · github · brave-search · playwright · context7 · *MCP servers, filtered per agent*
SERVING *inference runtime*	Ollama · OLLAMA_HOST=0.0.0.0 · vLLM — planned shared endpoint
MODELS *US & EU provenance*	Nemotron 3 Super · 123B · Devstral Small 2 · 24B · Gemma4 · 26B
HARDWARE *one node, on campus*	NVIDIA DGX Spark · GB10 Grace Blackwell · 128 GB unified memory · ARM64 · Ubuntu

LANGFUSE
self-hosted tracing, every layer

THE CONSTRAINT THAT SHAPED EVERYTHING

Texas DIR compliance requires US or EU provenance for every model and hosted service in the stack.

That single rule set the model roster, ruled out several popular open-weight families outright, and drove every hosting and gateway decision below the agent layer.

SKILLS LAYER

Teaching four agents the house style

Custom skills, not marketplace ones. Public skill packs assume frontier APIs, entangle non-compliant tooling, and carry audit findings we would have to own.

STEP 01

Agent node

PM, Dev, Test or Deploy asks for its operating instructions when the graph is built.

STEP 02

SkillLoader

Static per-agent mapping. No retrieval, no ranking, no embedding index — the same agent, the same skill.

STEP 03

Skill file

Markdown with lightweight YAML frontmatter, versioned with the code.

`--- name: pm agent: pm v: 1 ---`

STEP 04

Composed prompt

Inline prompt plus skill body, assembled once and handed to the local model.

NEVER MOVES

The inline prompt owns

The JSON output contract
Schema, required keys, and what an empty result is allowed to look like.

The competence floor
What the agent must refuse to guess at, and when it has to escalate.

The non-negotiables
Rules that must still hold if the skill file fails to load at all.

FREE TO EVOLVE

The skill file owns

Procedural elaboration
How to actually do the work, step by step, inside this codebase.

Project conventions
Naming, layout, and the house style the pipeline is expected to produce.

Worked examples
Concrete before-and-after cases a 24B model can pattern-match against.

THE DEDUPLICATION DECISION

The prose output-format section was deleted from the skill file. It restated a contract the inline prompt already enforced and charged tokens for it on every single call. The contract now lives in exactly one place, and the skill is free to change without breaking it.

CAMPUS AI INFRASTRUCTURE

Serving a university from eight boxes

A shared inference service for students, faculty and staff, built on hardware already sitting on campus — and around a compliance rule that removed most of the easy options.

DEMAND

Campus users

Students, faculty and staff across the university, not a lab of four people.

→

ACCESS

Campus VPN

luvpn.lamar.edu : 10443
LEA credentials + Duo 2FA. FortiClient ARM64, with openfortivpn as fallback.

→

INTERFACE

Open WebUI

One chat surface for the whole campus. Licensing tracked: v0.6.6+ adds a 50-user brand threshold.

→

ROUTING

LiteLLM gateway

LLM-aware load balancing. NGINX was rejected: it counts TCP connections, not GPU pressure.

→

EIGHT GB10 GRACE BLACKWELL NODES

Node 01 · DGX Spark vLLM · 128 GB unified	**Node 05 · Dell Pro Max** vLLM · 128 GB unified
Node 02 · DGX Spark vLLM · 128 GB unified	**Node 06 · Dell Pro Max** vLLM · 128 GB unified
Node 03 · DGX Spark vLLM · 128 GB unified	**Node 07 · Dell Pro Max** vLLM · 128 GB unified
Node 04 · DGX Spark vLLM · 128 GB unified	**Node 08 · Dell Pro Max** vLLM · 128 GB unified

Independent single-node workers, scaled horizontally. When one drops, the gateway routes around it.

ARCHITECTURE DECISIONS

Why it is built this way

Provenance before convenience
Texas DIR limits every model and hosted service to US or EU origin. Options were cut before testing.

A gateway, not a reverse proxy
NGINX balances TCP connections. LiteLLM balances what actually saturates here: the accelerator.

Two tracks, not one topology
Daily service and frontier research fail in different ways, so they get different network designs.

PHASE 3 · SEPARATE RESEARCH CAPABILITY

Unified multi-node cluster

The same eight boxes wired as one memory pool over a 400 Gbps RoCE fabric, for frontier-scale models — kept deliberately off the path students depend on daily.

OPERATING NOTES

Tailscale is blocked on the campus network, so all remote access runs through the university VPN. Eight independent workers degrade gracefully; one tightly-coupled cluster does not, which is exactly why the research fabric is kept off the service path.